c. CO_2 Removal

1) Absorption

Carbon dioxide formed in the Reformers and CO Converters is removed from the process stream to approximately 0.1% (dry basis) in the CO_2 Removal Section by absorption in a catalyzed solution of potassium carbonate commonly called the Catacarb Process. The difference between this system and the other commonly known hot potassium carbonate systems is the addition of a catalyst that will increase the absorbing rate of the carbonate solution as well as reducing corrosion. The general principle is that carbon dioxide is absorbed in a potassium carbonate (K_2CO_3) solution containing special catacarb catalysts and inhibitors while the hydrogen, nitrogen and traces of other gases not absorbed in the Catacarb solution pass out of the top of the Absorber for further treatment before it becomes

synthesis gas. The absorbed CO_2 is regenerated from the Catacarb solution in the CO_2 Regenerator (T-202). Approximately 90% of the heat required for the CO_2 Removal System is supplied by the heat content of the process gas coming from the Secondary CO Converter. The remaining heat requirement is supplied by 3 Kg/cm^2G steam in the Steam Reboiler (E-202). The carbon dioxide rich gas stream from the Secondary CO Converter passes through the Process Gas Desuperheater where the temperature of the process gas is controlled by injection of process condensate from the Catacarb Reboiler Separator (V-201) delivered by the Condensate Injection Pumps (P-206 A,B).

The saturated gas at 185°C flows to the Catacarb Reboiler (E-201) where heat is released to vaporize a flow of lean solution from the lower part of the CO_2 Regenerator (T-202). Water condensed from the process gas is separated in the Catacarb Reboiler Sèparator (V-201). A major part of the condensate from the Catacarb Reboiler Separator flows to the Degassing Columm (V-209) where CO_2 and other gases dissolved in the condansate is stripped off by air from Air Blower (K-201). The degassed condensate is delivered to Cooling Tower Unit by the Process Condensate Pump (P-208).

A minor part of the condensate is recycled to the Process Gas Desuperheater by the Condensate Injection Pump (P-206 A, B).

Cooled gas at 127°C enters the bottom of the CO_2 Absorber (T-201) where the carbon dioxide is contacted countercurrently with Catacarb solution in a two-stage absorption process. The bulk of the CO_2 in the gas stream is absorbed in the three lower packed sections of the tower by a semi-lean solution flow. Final absorption is performed in the top packed sections where a lean solution absorbs essentially all of the remaining CO_2 in the gas while simultaneously cooling the gas. Any carried

over moisture in the saturated purified gas is removed in the Absorber Knock Out Drum (V-204) before the process gas passes to the Methanator as the final synthesis gas purification. The Absorber Knock Out Drum protects the Methanator catalyst against Catacarb Solution carry-over during flooding or foaming conditions.

2) Regeneration

Catacarb solution accumulating in the bottom of the CO_2 Absorber, rich with absorbed CO_2, is commonly referred to as the rich solution stream. Leaving the bottom of the CO_2 Absorber at 112°C, the rich solution flows through a level control valve, passes through a Hydraulic Turbine (P-201HT) which recovers the power to assist the electric motor in driving the Semi-Lean Solution Pump (P-201A) and then flashes into the top of CO_2 Regenerator. The operating pressures in the CO_2 Absorber and CO_2 Regenerator are approximately 27 Kg/cm^2G and 0.7 Kg/cm^2G respectively at the bottom of each tower. The rich solution entering the CO_2 Regenerator is uniformly distributed down through the center section of tower where the CO_2 in solution is stripped off by a rising flow of steam. A stream of partially regenerated Catacarb solution, commonly referred to as the semi-lean solution stream, is drawn off at the intermediate section of the Regenerator to the Semi-Lean Solution Pump (P-201 A, B). The semi-lean solution stream is pumped up to the semi-lean section of the CO_2 Absorber through the Semi-Lean Solution Cooler (E-207) where the semi-lean solution is cooled down from 114°C to 105°C.

At the bottom of the CO_2 Regenerator the solution is almost completely regenerated by the two Reboilers producing the very lean solution.

The required heat of regeneration is supplied by the heat of process gas at the Catacarb Reboiler (E-201) and by 3 Kg/cm^{2}G steam in the Steam Reboiler (E-202). The lean solution from the CO_2 Regenerator is cooled to 80°C in the BFW/Lean Solution Exchanger (E-204) before pumped into the top of the CO_2 Absorber by the Lean Solution Pump (P201 AB). The temperature of the lean solution to the Absorber is controlled with by-passing of boiler feed water around the Exchanger. Similarly the semi-lean solution boosted up by the Semi-Lean Solution Pump (P-202AB) is cooled to 105°C in the Semi-Lean Solution Cooler (E-207). The flow rate of both the semi-lean and lean solutions are controlled into the CO_2 Absorber. A side stream of lean solution is continuously filtered through the Lean Solution Filter (F-201) before it rejoins the main flow of solution entering the Absorber. The anti-foam chemical is injected into the rich or, semi-lean and/or lean solution by the Anti-Foam Injection Pump (P-205AB). The Inhibitor in the solution is maintained in an active state by the introduction of a small quantity of air from the Process Air Compressor into the rich solution line at the discharge of the Hydraulic Turbine.

The water balance of the Catacarb solution is maintained by controlling the reflux of acid condensate from the Acid Gas Separator (V-203) to the top of the Regenerator where four wash trays are used to eliminate carbonate carryover from the Regenerator. Excess acid condensate is sent to Degassing Column by the Reflux Pump (P-203AB). Most of acid condensate is sent to the wash tray. Wash condensate from the Regenerator wash section is returned to the inlet header piping for the Catacarb Reboiler and Steam Reboiler where it is vaporized.

By-product CO_2 leaves the top of the Regenerator at 97°C and is cooled by cooling water in the Overhead Condenser (E-205AB) to 50°C. After condensed water is removed in the Acid Gas Separator (V-203) excess CO_2 not transferred to the Urea Unit is vented to the atmosphere by pressure control. (PIC 203)

A Catacarb Storage Tank (V-205) is provided to store Catacarb solution in and a Solvent Transfer Pump (P-204) and Sump Tank (V-207) to add solution to the system as necessary.

d. Methanation

The Methanator (R-106) converts the residual carbon monoxide or the unabsorbed carbon dioxide to methane, for carbon oxides are poisonous to the ammonia synthesis catalyst.

The decarbonated gas from the CO_2 Absorber is preheated by the Methanator effluent in the Methanation Heat Exchanger (E-106) and further heated up to 310°C by the Primary CO Converter effluent in the Methanation Preheater (E-107) before feeding to the Methanator. The Methanator feed temperature is controlled by TIC-124. The make-up gas passes downward through the Methanator catalyst bed and out through the tube side of the Methanation Heat Exchanger (E-106).

The Catalyst promotes the reaction of carbon oxides with hydrogen to produce methane and water leaving a maximum of 10 ppm of residual carbon oxides. The Methanator effluent now has the following composition in mol % on a dry basis:

Hydrogen	74.06%
Nitrogen	24.69%
Methane	0.85%
Argon	0.4%
Carbon Oxides	10 ppm

This is the design base composition of the make-up Synthesis gas.

As final preparation enroute to the compression section, the process gas is cooled in the Syn. Gas Precooler (E-301).

Any condensed moisture is removed in the Syn. Gas Precooler Separator (V-301) before passing on to the Synthesis Gas Compressor (K-301) in preparation for ammonia production.